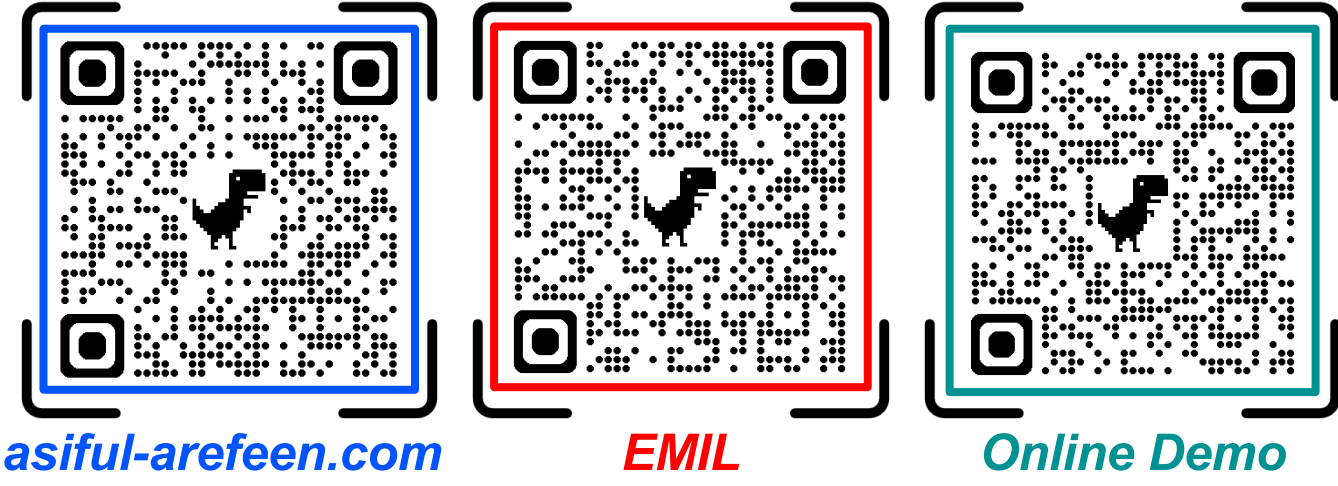


# Glucoguide: A Retrospective Counterfactual Decision Support to Address Stakeholder Needs

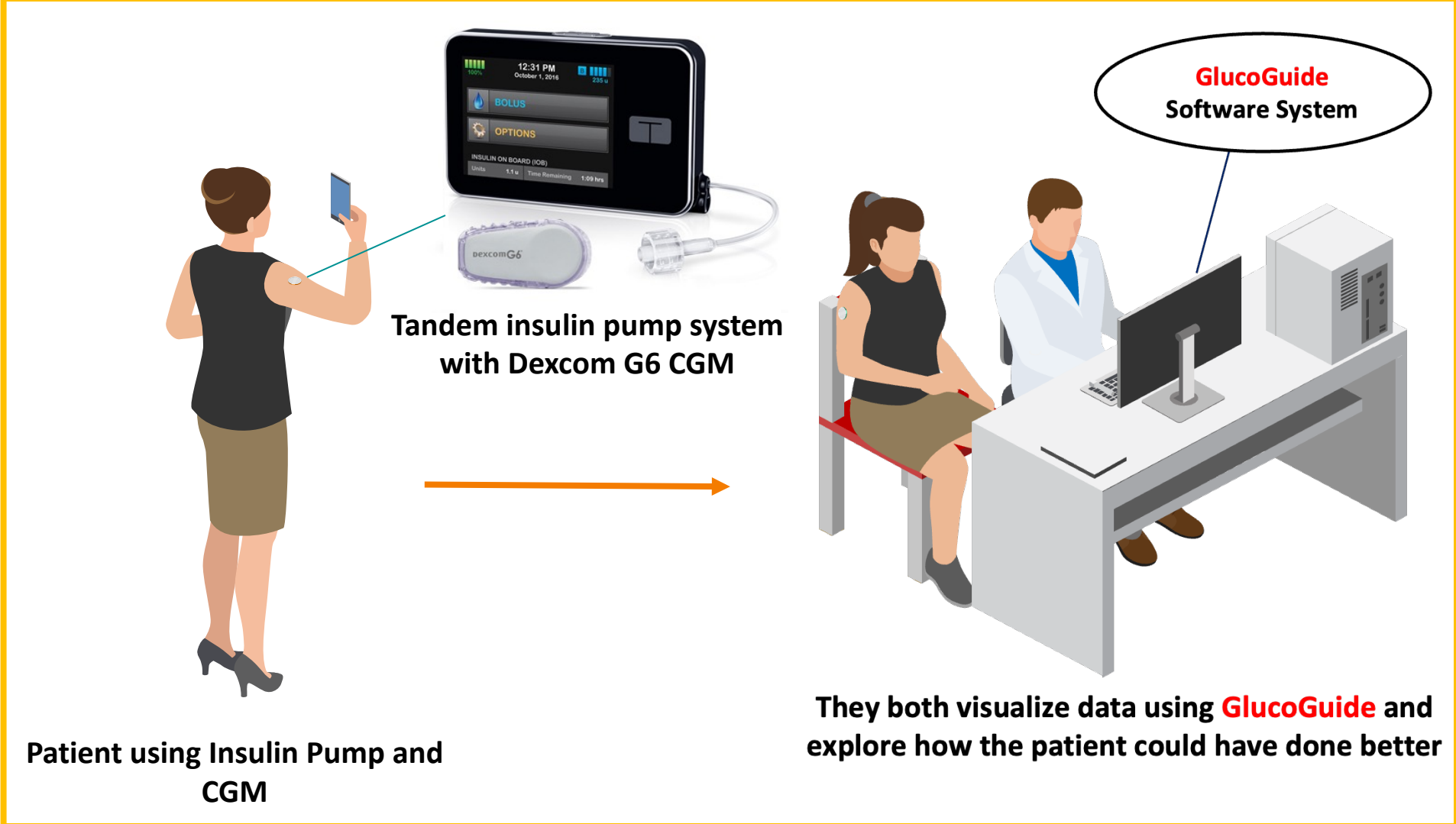
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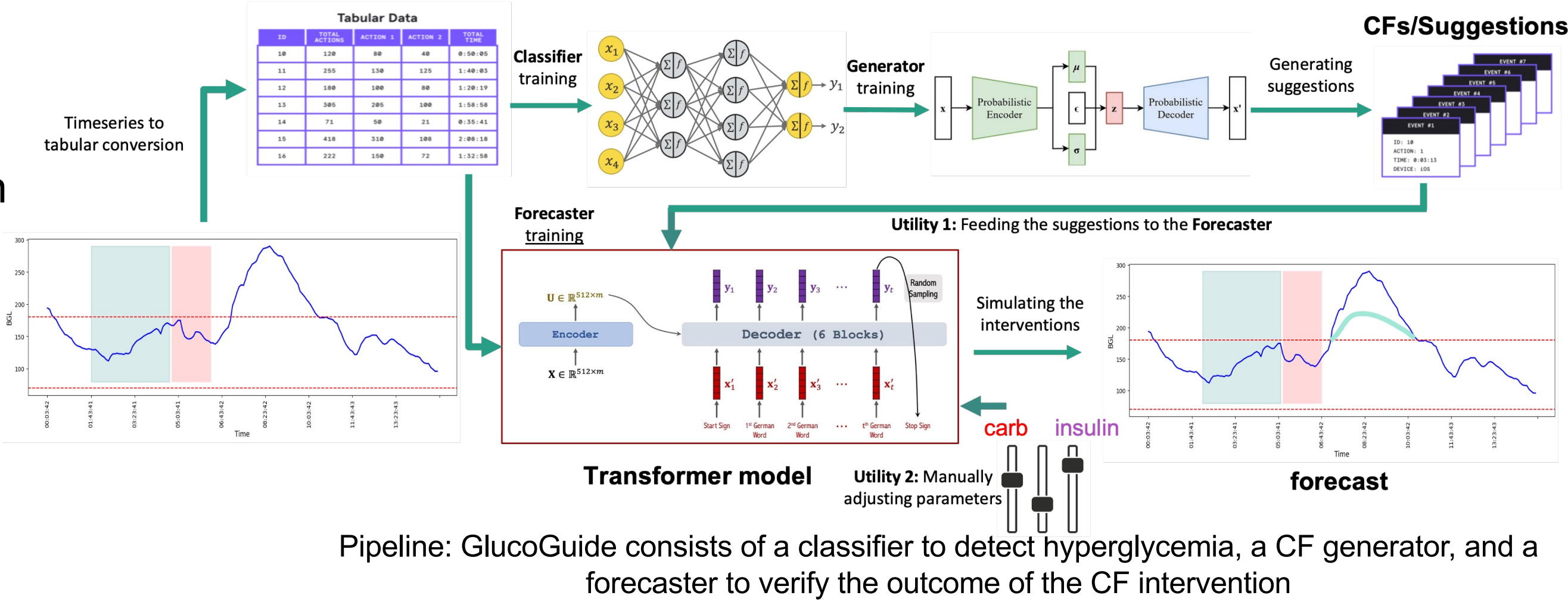
## INTRODUCTION

Counterfactual explanations (CFs) refer to minimal alterations to user attributes fed into a model that result in a better predicted outcome. We present **Glucoguide**, a software to assist endocrinologists in retrospective clinical decision making while consulting with patients with Type 1 Diabetes (T1D) who use Automated Insulin Delivery (AID) and Continuous Glucose Monitoring (CGM) technology. GlucoGuide can help patients understand how they could have improved insulin dosing adjustments to achieve better glycemic control in situations where they experienced hyperglycemia.



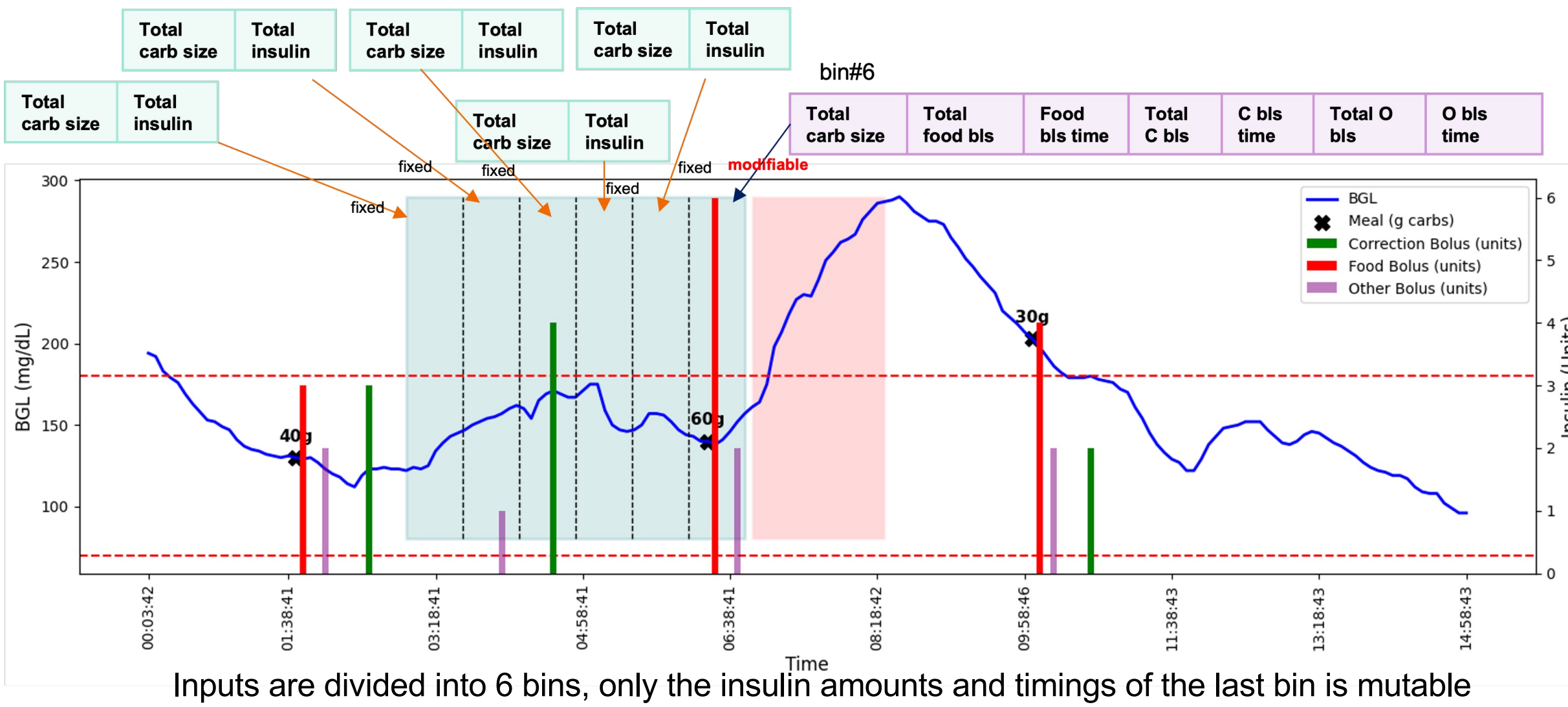
## AIMS

1. Develop a web portal to use in clinical setting by doctors for retrospective intervention design while they consult with patients with T1D
2. The web portal will generate interventions by modifying insulin amounts and times, do a sanity check of the interventions with a glycemic response forecaster
3. GlucoGuide will be tested by a cohort of diabetes experts



## DATASET

- **Participants:** 38 patients with Type 1 Diabetes monitored in free-living condition
- **Devices:** **Dexcom G6 CGM** to capture blood glucose every 5 minutes, **Tandem Insulin Pump** for releasing insulin when needed
- **Study time:** Each patient was monitored for 26 days on average
- **Features:** Data includes CGM signals from insulin logs, carb sizes, and device modes (regular/sleep/exercise) from the insulin Pump.



## METHODS

1. The continuous data from the insulin pump is **processed** into tabular format
2. A Dual Attention Network (DANet) **classifier** is trained that can detect hyperglycemia beforehand
3. A conditional Variational Autoencoder (cVAE) **generator** is trained to generate CF suggestions. When fed with a context that leads to hyperglycemia, the role of the generator is to generate a CF recommendation by modifying insulin intakes such that the classifier predicts it as a normoglycemic sample
4. The **forecaster** is built with a sequence-to-sequence transformer. It helps providers verify if the recommendations lead to positive outcomes within the desired glucose range. Providers can also change context attributes, design interventions themselves and visualize the outcomes using the forecaster

## RESULTS

### Performance

The classifier is **84.12%** accurate, the CF generator has a validity of **94.29%** while the forecaster has an RMSE or **7.24 mg/dL**. We have also developed the interactive web-application and getting the interventions and the application is being evaluated by diabetes experts

| Classifier Performance   |                |                  |
|--------------------------|----------------|------------------|
| Accuracy (%)             | F1-score       | AUC              |
| 84.12%                   | 83.91%         | 0.9063           |
| CF Generator Performance |                |                  |
| Validity (%)             | $L_2$ Distance | Plausibility (%) |
| 94.29%                   | 63.86          | 98.22%           |
| Forecaster Performance   |                |                  |
| RMSE (mg/dL)             | MAE (mg/dL)    | $R^2$            |
| 7.24                     | 5.02           | 0.976            |

The above table explain the performance of the individual components of GlucoGuide

